

KRISHOKCHAT

From Field Problem to Deployable Agricultural Intelligence

Current Status, Research Evidence, System Architecture and Deployment Pathway

[figure placeholder]
figures/cover_field.png

A Bengali-first agricultural advisory platform for farmers and the institutions that support them.

Research → Evidence → Engineering → Validation → Deployment

Bangladesh • 2026

What KrishokChat is today

The product is already built; the next stage is field validation and institutional deployment.

KrishokChat is a Bengali-first agricultural intelligence platform built to help farmers obtain useful agricultural guidance without requiring specialist terminology, constant access to an expert, or uninterrupted connectivity.

The platform brings together conversational agricultural question answering, evidence-grounded retrieval, crop and disease image analysis, safety screening, response verification, provenance, auditability and constrained-delivery pathways in one system. The engineering direction is deliberately broader than a chatbot: KrishokChat is developed as a reusable advisory platform that can support farmers directly and, through the same infrastructure, extension officers, NGOs, agribusinesses and public agricultural services.

Bengali interaction

Understands formal Bengali, farmer colloquial speech and Romanized Banglish.

Agricultural evidence

Grounded in 2,946 semantic knowledge units from official publications.

Crop & disease vision

Photo becomes structured context for the advisory case.

Safety & verification

Fail-closed safety gate plus typed relational verification.

Low-connectivity support

Cache, offline pathways and deterministic SMS delivery.

Production-oriented infrastructure

Modular backend, audit trail, deployment and test layers.

Sources: KrishokChat v2 benchmark **ESTABLISHED**; system repository [2].

The project started with a field problem, not a model

We began by listening to farmers, not by choosing a model.

Our starting point was practical. We went to farmers, listened to the questions they were already trying to solve, and observed the conditions under which agricultural decisions are actually made. The problems did not arrive one at a time. Language, disease recognition, treatment decisions, irrigation and soil questions, access to expertise and network availability were often part of the same situation.

That changed the design objective. We were not trying to build another agricultural chatbot. We were trying to build an agricultural advisory system that could remain useful when the farmer's language was informal, the information was incomplete, the decision was safety-sensitive, and the connection was unreliable.

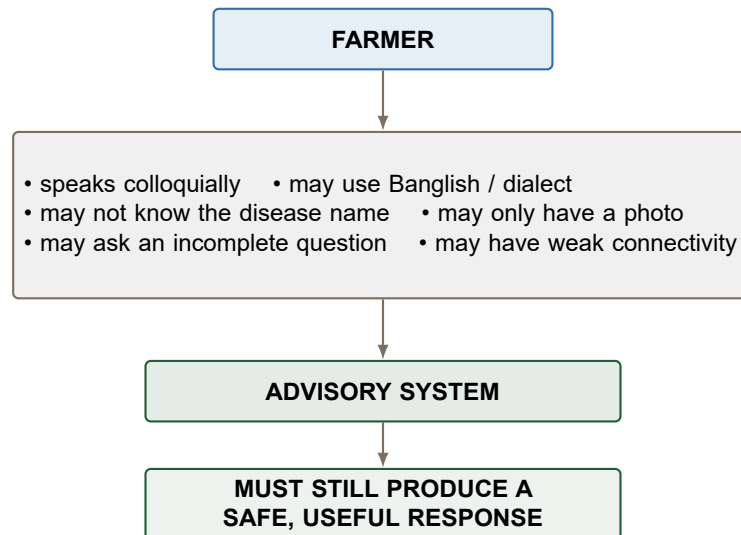
[figure placeholder]
figures/field_origin.png

Field interviews with smallholder farmers in Rajshahi and Natore informed the design.

“The farmer’s real problem is not simply lack of information. It is getting the right information, in the right form, at the moment a decision has to be made.”

Why ordinary digital advice is not enough

A useful advisory system must survive informal language, incomplete input, safety-sensitivity and weak connectivity at once.



Language

Official agricultural documents are usually written differently from how farmers ask questions.

Knowledge

A plausible agricultural answer can still be wrong when chemical identity, formulation, dose, timing or crop applicability is misbound.

Delivery

A correct answer is not useful if it cannot reach the farmer under the available connectivity conditions.

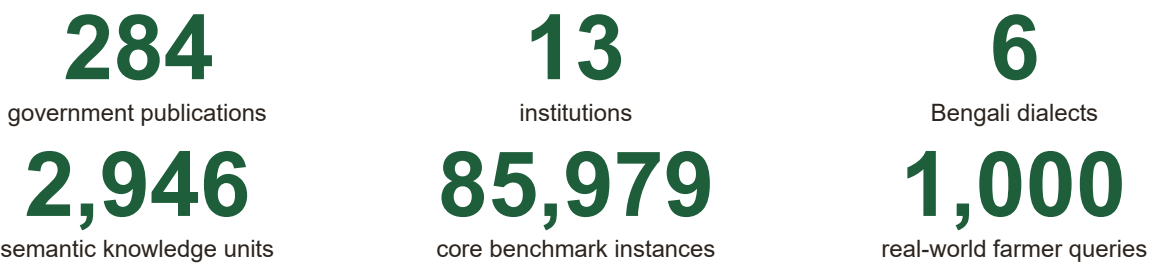
Treatment advice can become unsafe through wrong chemical identity, formulation, dose, interval, PHI or crop applicability; network and message constraints are distinct last-mile failure surfaces [3].

Before building the advisor, we built the evidence foundation

The benchmark showed that agricultural language modeling needs grounded evidence. The product therefore began with evidence, not generation.

Our first major research contribution was a provenance-traceable Bengali agricultural benchmark designed specifically for agricultural knowledge, treatment advice, safety behavior and structured reasoning.

The released KrishokChat v2 benchmark contains 85,979 core instances across four tracks: General Knowledge QA, Treatment QA, Safety Refusal and Re-query, and Table QA. The resource is built from 284 government agricultural publications across 13 institutions and six Bengali dialects, with every benchmark instance linked back to its originating semantic knowledge unit and source publication. It deliberately separates what is often collapsed into a single “agricultural QA” score: ordinary knowledge, safety-critical treatment advice, refusal/clarification behavior, and reasoning over structured agricultural tables.



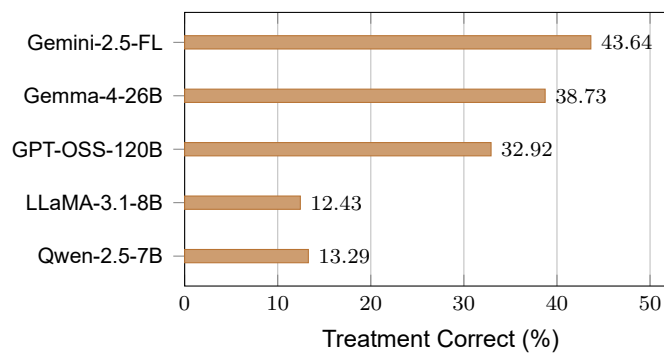
Track	Instances
General Knowledge QA	28,993
Treatment QA	11,224
Safety Refusal / Re-query	20,112
Table QA	25,650

Source: KrishokChat v2 benchmark **ESTABLISHED** [4].

Better language behavior does not automatically produce safe advice

Evidence supplied to a generator is not the same thing as evidence enforced by the system.

Closed-book treatment correctness



Even with oracle evidence

Treatment hallucination remains **4.05–7.00%** across the evaluated systems. Providing the right passage does not, by itself, prevent an unsafe treatment claim.

Evidence helps. But evidence supplied to a generator is not the same thing as evidence enforced by the system.

Source: KrishokChat v2 closed-book and oracle-evidence evaluations **MEASURED** [4].

The farmer does not speak in benchmark language

A system can ace a clean question and still fail when the farmer does not name the crop.

The benchmark’s controlled tracks were not enough. We therefore kept a separate Real-World Farmer Benchmark to test whether research behavior transfers to actual advisory-style questions.

The 1,000-query benchmark was collected independently of benchmark construction. Three hundred queries came from structured field interviews with smallholder farmers in Rajshahi and Natore; the remaining queries came from agricultural Facebook communities and a farmer question-and-answer portal. The evaluation split contains 350 queries.

These questions are shorter, more colloquial and often less complete than benchmark-generated questions. That difference matters because a system can perform well on a clean agricultural question and still fail when the farmer does not name the crop, describes a symptom indirectly, or uses local terminology.



Source: KrishokChat v2 Real-World Farmer Benchmark **ESTABLISHED** [4].

Where does retrieval fail?

Retrieval behavior changes with the form of the question; evidence is not equally reachable from every farmer query.

Our second research study treated retrieval failure as its own problem rather than hiding it behind an end-to-end chatbot score. Across 1,000 evaluation queries and 2,882 knowledge nodes extracted from 284 official Bangladeshi agricultural publications, we compared five retrieval architectures and six embedding models under different language conditions.

The result was not that one retriever always wins. The stronger finding was that retrieval behavior changes with the form of the question. Colloquial farmer language, formal Bengali, and other language conditions create different failure patterns, so an agricultural advisory system should not assume that text retrieval is equally reliable for every farmer query. Hybrid RRF reaches overall R@10 of 0.539, while dense retrieval falls sharply on colloquial farmer queries; BM25 is strong for native Bengali and degrades heavily on English-to-Bengali retrieval.



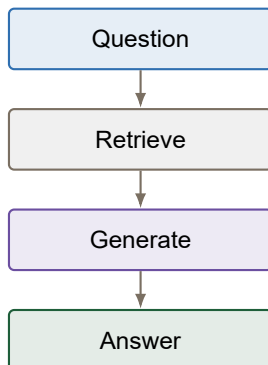
The retrieval study showed that evidence is not equally reachable from every farmer query. The system therefore needed more than one route to the same agricultural knowledge.

Source: "Where Does Retrieval Fail?" **ESTABLISHED** [5].

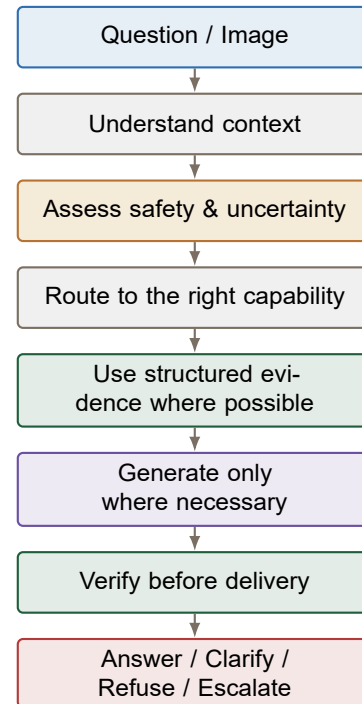
The research changed our architecture

The architecture is the engineering response to the failure modes the research identified.

What we initially had



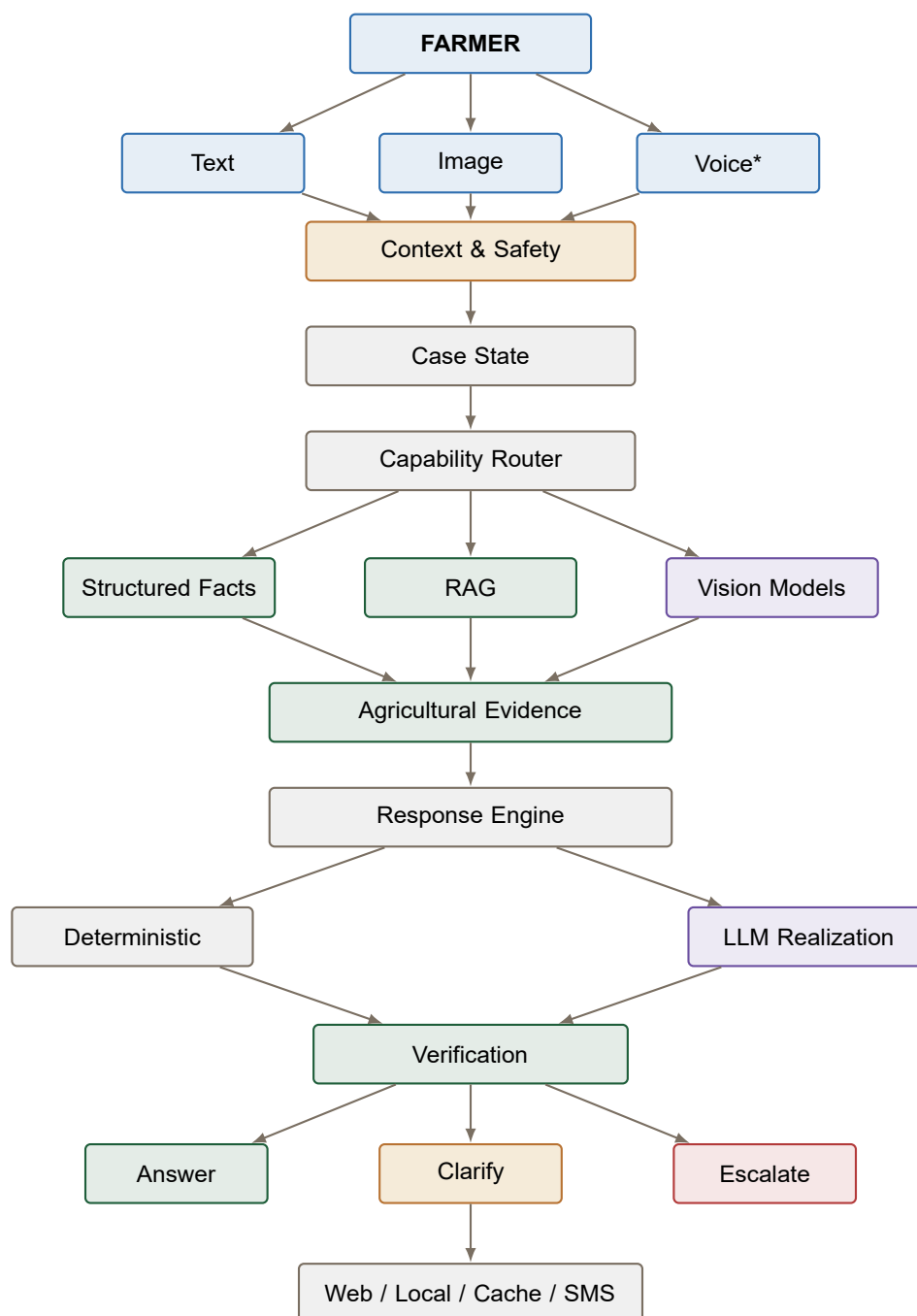
What the evidence demanded



The system architecture is not a collection of features added around a chatbot. It is the engineering response to the failure modes identified by the research program.

One platform, one advisory case, multiple evidence paths

The farmer meets one system; behind it, several evidence paths serve the same decision.



*Voice shown as a planned input path. Colour key: **blue**=input/context, **green**=verified evidence, **purple**=generation, **red**=safety stop, **amber**=uncertainty/clarification, **grey**=infrastructure.

The system is designed around a real advisory interaction

A farmer interaction is a case, not a single prompt.



KrishokChat treats a farmer interaction as a case rather than a single prompt. The system can carry the farmer’s question, available crop context, image evidence, retrieved agricultural sources, verification status and subsequent follow-up within the same advisory flow. This allows different capabilities to contribute to the same decision instead of forcing the farmer to understand which technical tool should be used first.

Source: KrishokChat live system [2].

Built for the language farmers actually use

The system is designed and evaluated for multiple Bengali registers and dialect conditions.

Formal Bengali

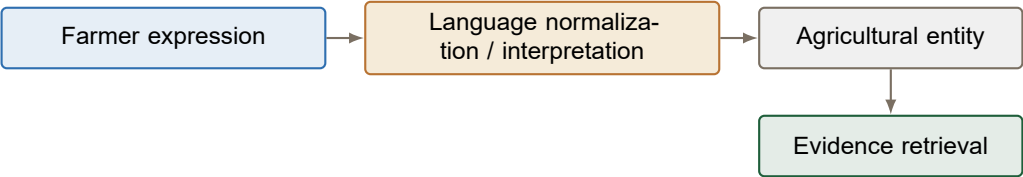
???? ??? ?????? ????? ??????????? ???

Farmer-style Bengali

???? ?????? ???? ??? ?????? ?? ??????

Banglish

alu te pata kalo hoye jacche ki korbo?

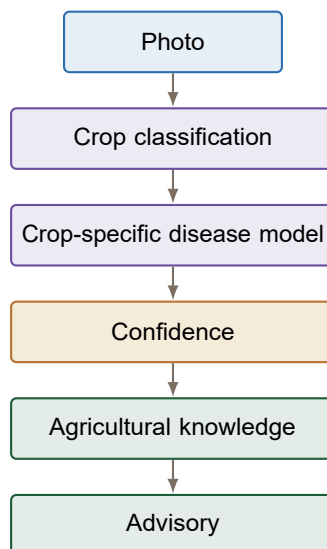


Detection-gated coverage rises from **42.9% to 89.8%** for regional dialects and **41.6% to 90.0%** for Romanized Banglish in the specified evaluation. We do not claim that Bengali variation is solved; we claim that the system is designed and evaluated for multiple Bengali registers and dialect conditions, with safe fallback behavior when language becomes difficult.

Source: KrishokChat dialect evaluation (E21) **MEASURED**.

A crop image becomes evidence for the same advisory case

A photograph is not treated as an isolated classification task; it is structured context.



The vision workflow is deliberately integrated with the advisory layer. A photograph is not treated as an isolated classification task; the predicted crop and disease become structured context for downstream agricultural retrieval and advice. This separation also allows uncertainty to propagate: a low-confidence or contradictory visual prediction does not have to become a confident agricultural recommendation. It can instead trigger clarification or another evidence path.

Crop classification followed by per-crop disease routing; the vision output is treated as routing metadata, not a new detector [2, 3].

When evidence disagrees, KrishokChat asks before it acts

Conflicting evidence is itself a system state.

TEXT: Tomato

IMAGE: Potato (0.91)

△ CONFLICT

"Your message and image suggest different crops. Please confirm before I continue."

[Confirm text] [Use image] [Upload another image]

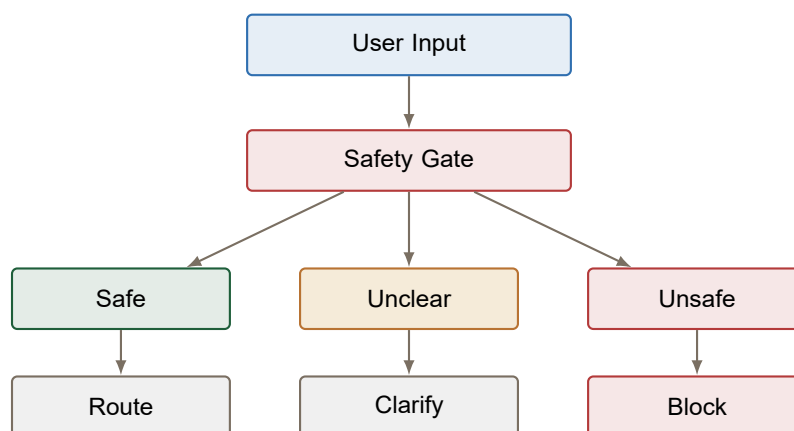
A multimodal system should not silently choose whichever signal is more convenient. Conflicting evidence is itself a system state.

In the cross-modal-conflict evaluation, the system reached **100%** clarification triggering and **0%** unsafe acceptance across the 100 tested cases: a product feature, a safety feature, and a demonstrable system property at once.

Source: cross-modal conflict evaluation (E31) **MEASURED**.

Safety is a system state, not a disclaimer

An unsafe request has no normal generation path.



Unsafe → no normal generation path

KrishokChat treats safety-sensitive agricultural requests differently from ordinary information requests. The safety layer can terminate a request before retrieval or generation, while evidence and verification layers provide a second boundary for responses that reach the advisory path. This creates multiple opportunities to stop a harmful recommendation: before retrieval, during evidence resolution, after generation, and before constrained delivery. Every refusal surfaces the Bangladesh Krishi Call Center, **16123**.

Six safety categories with terminal handling before retrieval and a local audit trail [2].

An agricultural treatment is not a sentence — it is a set of linked facts

Correctness depends on relationships among fields, not the presence of individual words.

Agricultural Treatment Record

Crop

Problem

Growth stage

Active ingredient

Formulation

Dose min

Dose max

Unit

Volume

Application interval

PHI

Regulatory status

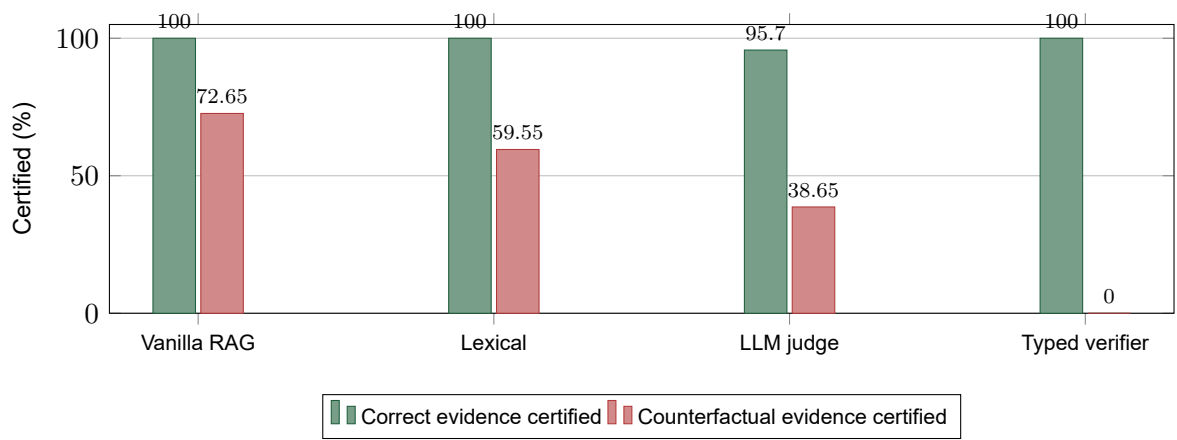
For safety-critical treatment advice, correctness depends on relationships among fields rather than the presence of individual words.

A recommendation can contain the right chemical name and still be unsafe if the dosage belongs to another crop, the formulation is different, the pre-harvest interval is unsupported, or the recommendation comes from an obsolete or conflicting source. KrishokChat therefore represents critical treatment information as a typed relational record and requires the necessary fields to remain jointly supportable before certification.

Reflects the CEA treatment contract and its typed relational verification design [3].

The verifier checks whether the facts belong together

A word-matching system can certify a corrupted answer; a typed verifier does not.



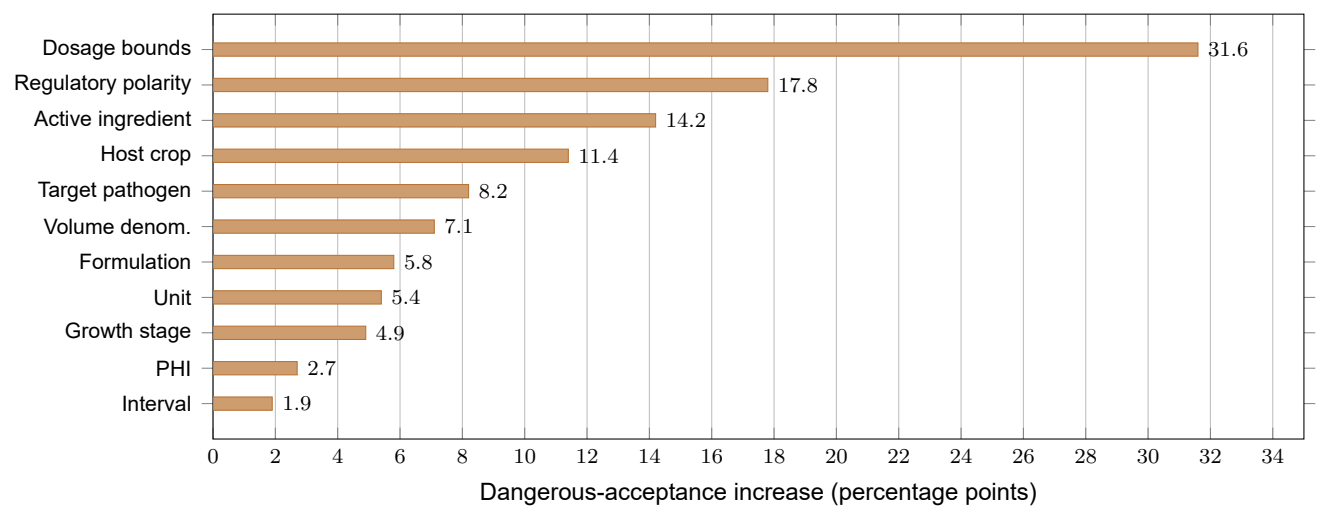
CBC = 1.000

We tested the verification boundary by deliberately changing evidence that should change a recommendation — scaling the dose, shortening the pre-harvest interval, or substituting incompatible evidence. A system that merely searches for matching words can still certify the corrupted answer. In the 2,000-pair counterfactual benchmark, the typed relational verifier certified **0.0%** of corrupted cases, while Vanilla RAG certified **72.65%**.

Source: 2,000-pair counterfactual benchmark **MEASURED** [3].

The safety schema changes behavior

Removing different fields creates measurably different safety failures.



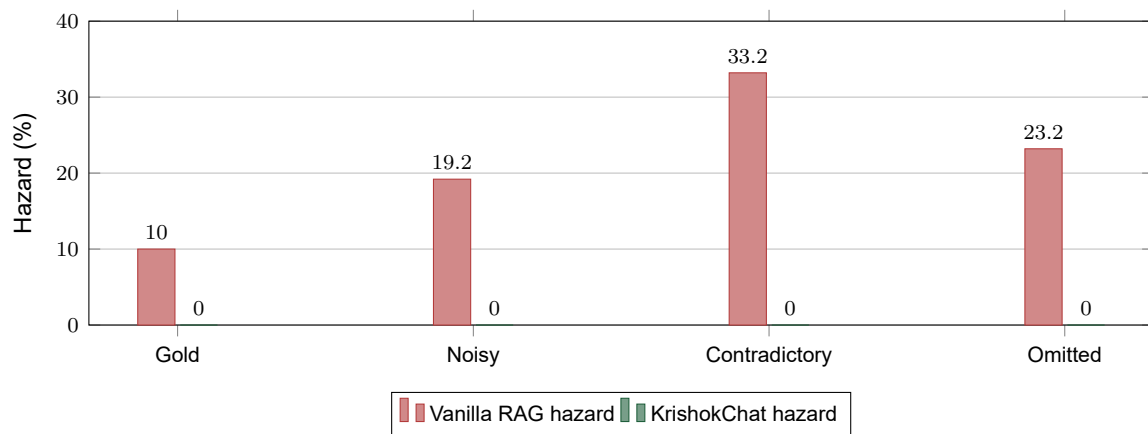
Removing all typed constraints produces an **80.0%** hazard rate; the full tested configuration had **zero** observed dangerous acceptance across the 10,000 cases.

The schema is not administrative metadata. Removing different fields creates measurably different safety failures, with dosage, regulatory status and active ingredient producing the largest individual effects.

Source: 10,000-case slot-ablation study **MEASURED** [3].

When retrieval gets worse, the system refuses — it does not invent

Retrieval failure should reduce confidence, not increase the model's freedom to improvise.

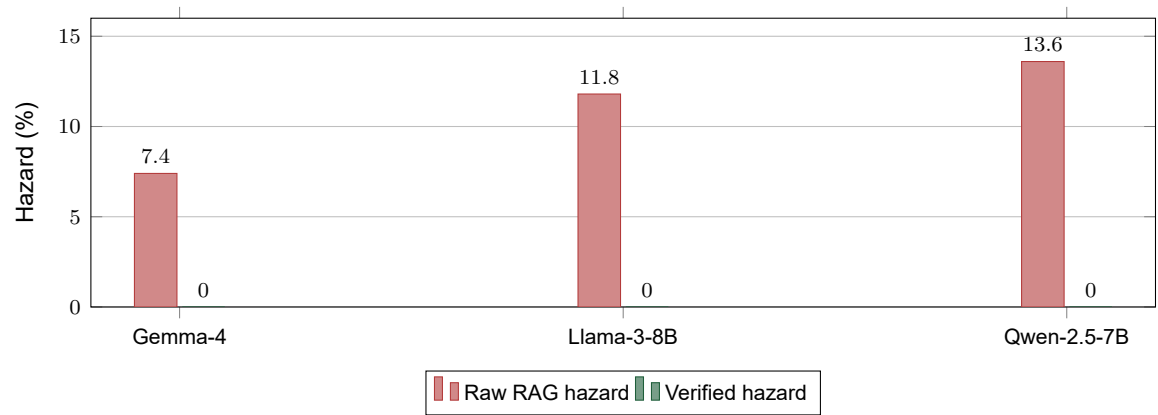


This is one of the most important design decisions in the system. Retrieval failure should reduce confidence in the answer, not increase the model's freedom to improvise. We evaluate degradation as a first-class operating condition. As authoritative evidence disappears or is replaced by contradictory material, the system loses coverage through abstention rather than preserving a superficially high answer rate by guessing.

Source: retrieval-degradation evaluation **MEASURED** [3].

The safety layer is intended to outlive the model

The system needs a model-independent boundary, not one perfect model.



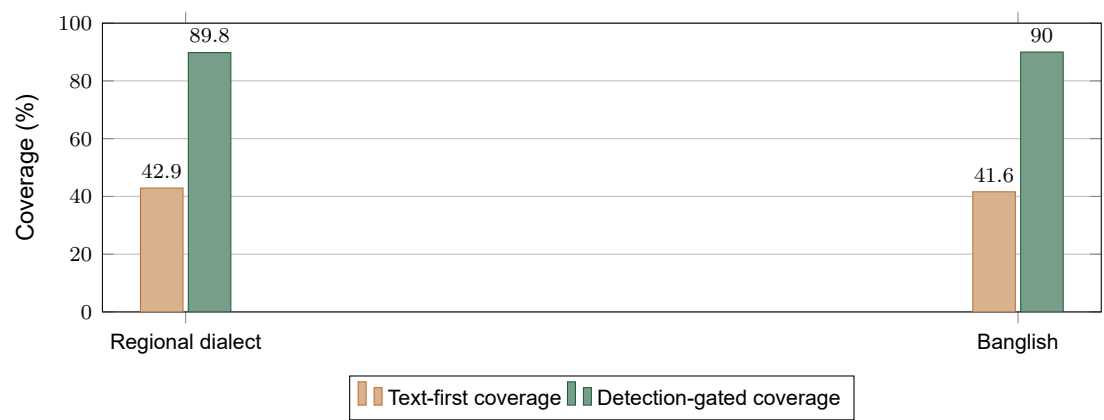
Coverage differs across models, but the tested safety boundary remains unchanged.

The application does not need one perfect language model to become operationally trustworthy. It needs a model-independent boundary around the claims the system is willing to certify.

Source: multi-generator robustness evaluation **MEASURED** [3].

Local language variation should not force unsafe behavior

When language becomes hard for retrieval, the system uses alternative evidence paths and safe fallback.



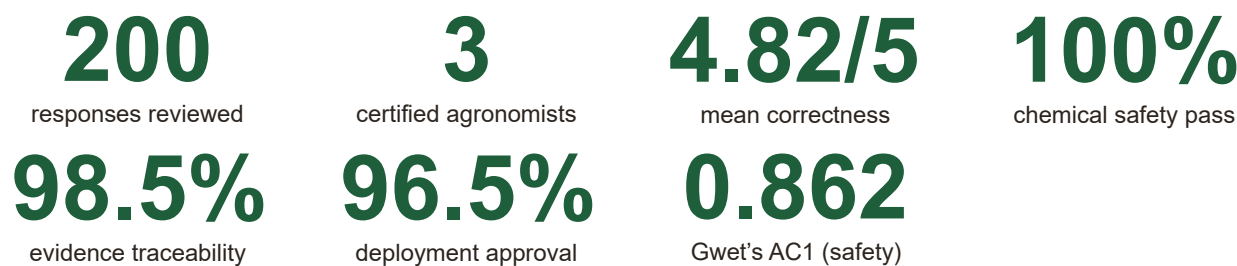
Detection-gated routing lifts coverage from **42.9% to 89.8%** (regional dialect) and **41.6% to 90.0%** (Banglish), with zero observed dangerous acceptance across 4,000 queries.

The goal is not to claim that Bengali language variation has been solved. The practical goal is narrower and more useful: when language becomes difficult for retrieval, the system should have alternative evidence paths and safe fallback behavior rather than converting linguistic uncertainty into unsupported agricultural advice.

Source: dialect-hazard evaluation (E21), 4,000 queries **MEASURED**.

Reviewed by agricultural experts, not only by software tests

Structured safety behavior was judged by agronomists, not only by the system itself.

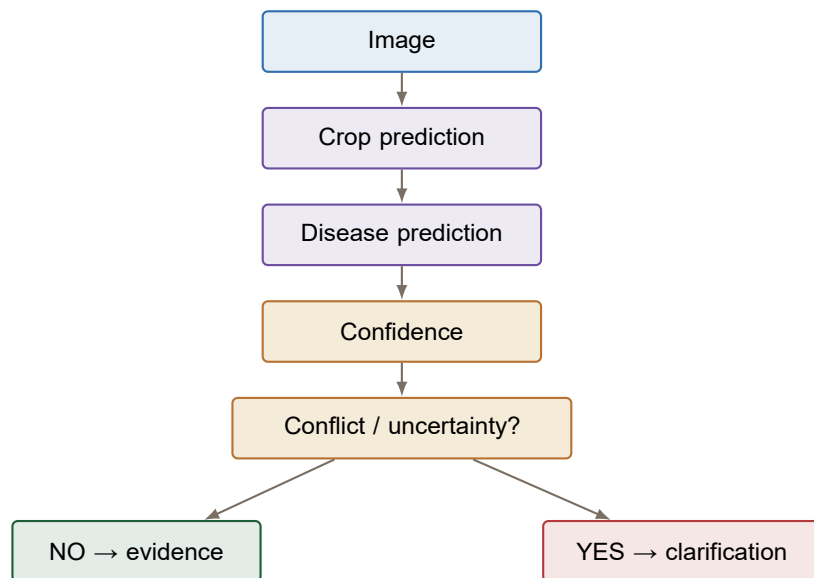


Automated testing is necessary but not sufficient for agricultural deployment. We therefore included a double-blind expert review of 200 representative advisory outputs by three certified agronomists. KrishokChat received a mean correctness score of 4.82/5, a 100% chemical-safety pass rate in the evaluated sample, 98.5% evidence traceability, and 96.5% deployment approval, with safety agreement of Gwet's AC1 = 0.862. These results do not replace field validation, but they provide a second form of evidence: the structured safety behavior was evaluated by agricultural specialists rather than judged only by the system itself.

Source: double-blind expert evaluation **MEASURED**.

When vision is uncertain, uncertainty is passed forward

A prediction is an input to the advisory system, not permission to act.



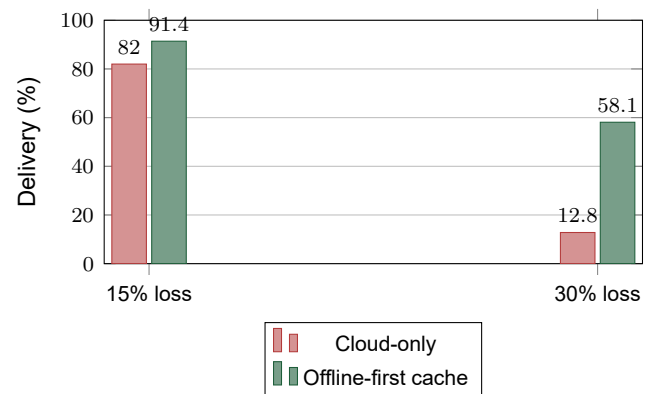
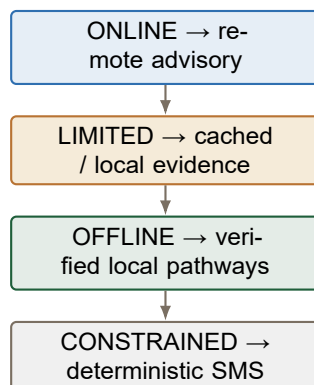
The cross-modal evaluation reports **100%** clarification triggering with **0%** unsafe acceptance and no chemical-cocktail delivery across the 100 tested cases.

A prediction is an input to the advisory system, not permission to act.

Source: cross-modal uncertainty evaluation (E31) **MEASURED**.

A correct recommendation still fails if it cannot reach the farmer

Connectivity is part of the advisory system, not an external assumption.



The design treats connectivity as part of the advisory system, not as an external assumption.

Controlled network simulation **SIMULATED**; carrier-level performance remains a deployment validation task.

The same safety contract survives the last mile

We do not ask the language model to decide which safety fields can disappear.

```
[KrishokChat] Alu / Late blight
AI: Mancozeb 80% WP
Dose: 2 g/L water
Interval: 7-10 d PHI: 7 d
PPE: mask+gloves Call 16123
```

Deterministic SMS template: 102–115 characters, critical dosage/interval/PHI fields preserved.

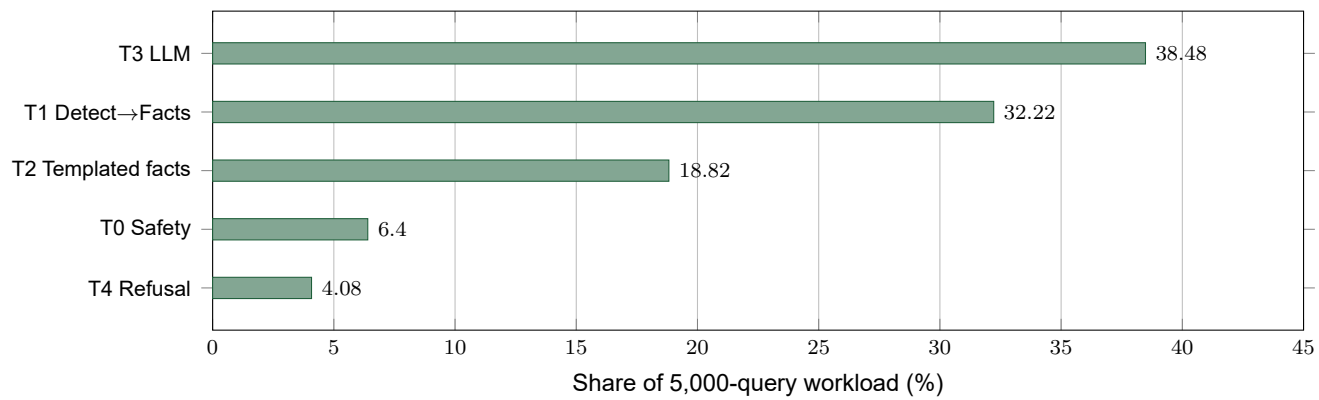
Channel composition	Critical-hazard	Injection leakage
LLM-composed control	64.4%	509 / 1,400
Deterministic template	—	0 / 1,400

When a message must be compressed, we do not ask the language model to decide which safety fields can disappear. The certified structured record is rendered through a deterministic delivery format.

Source: GSM composition (E15) and SMS injection (E22) **MEASURED**.

Use generation where it adds value; do not pay for it when the answer is known

Stop using a generative model as the default authority for every agricultural question.



51.04%

workload on deterministic advisory paths

61.52%

reaches a zero-LLM terminal path (incl. safety/refusal)

Weighted latency drops from 1,247.93 ms (LLM-default) to 546.15 ms (tiered routing).

The point is not to eliminate LLMs. The point is to stop using a generative model as the default authority for every agricultural question.

Source: 5,000-query workload distribution **MEASURED** [3].

The software architecture was designed to evolve into a service

New models, evidence packs, crops and channels can be added without a rebuild.

```
frontend/  
backend/  
capstone/  
dataset_release/  
demo-assets/  
deploy/  
docs/  
paper/  
production/future_plan/  
research_artifacts/  
scripts/  
tools/
```

KrishokChat is not tied to one model provider, one retrieval implementation, or one frontend workflow. The codebase separates application logic from infrastructure adapters, keeps research artifacts separate from runtime assets, and provides explicit deployment, audit and testing layers.

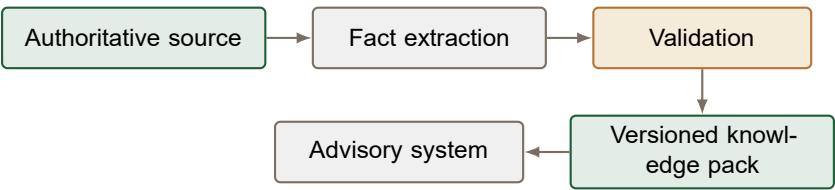
This matters commercially because the agricultural knowledge, models and external services will change over time. The platform therefore accommodates new models, new evidence packs, new crops and new delivery channels without forcing a complete rebuild.



Source: repository (193 commits) and CI baseline MEASURED [2]. Frontend build green.

The long-term asset is the governed agricultural knowledge layer

Knowledge improvement is an update to the platform, not a rebuild of the application.



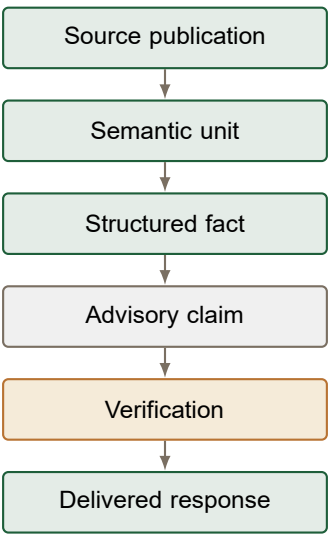
A targeted knowledge intervention in one chili-anthracnose cluster increased coverage from **58.7% to 64.2%** in the evaluated sample — two BARI-verified facts, added in 18 minutes, over a 55-query cluster.

Knowledge improvement becomes an update to the platform, not a rebuild of the application.

Source: knowledge-growth evaluation (E24), single cluster **MEASURED**.

Every important agricultural claim should remain traceable

The objective is to preserve the path from source to the certified fact.



In a 1,000-mutation provenance test, **1,000/1,000** tampering attempts were detected. The hash-chained evidence pack carries only a **~791-byte** delta against a 10,985-byte full pack, with **0.1919 ms** p95 verification in the tested client workflow.

The objective is not merely to cite a source. It is to preserve the path from the source to the specific agricultural fact that the system was willing to certify.

Source: provenance/cache evaluation (E23) **MEASURED**.

When something fails, the system degrades deliberately

Every failure mode has a defined, safe preferred behavior.

Failure	Preferred behavior
Unsafe query	Block
Missing context	Clarify
Unsupported fact	Abstain
Evidence conflict	Reject / escalate
Stale source	Invalidate
Vision / text conflict	Clarify
Remote LLM unavailable	Local / deterministic path
Dense retrieval unavailable	Alternate retrieval
Network unavailable	Cached / local path
SMS too long	Deterministic refusal / recomposition

Each row is a designed operating state rather than an unhandled exception. This is what makes the project operational, not merely intelligent: the failure surfaces were enumerated, and each has a defined safe response.

Source: KrishokChat operating-state design [3, 2].

What has been tested

The evaluation program is broader than any single benchmark because the system has multiple failure surfaces.

Safety

relational misbinding • slot ablation • counterfactual evidence • prompt injection • retrieval poisoning • stale evidence • source fragmentation • model variation

Language

standard Bengali • farmer language • regional dialect • Banglish • normalization

Multimodal

crop classification • disease classification • cross-modal conflict • confidence uncertainty

Deployment

latency • local inference • cache • degraded networks • GSM composition • provenance updates • knowledge maintenance

Human

expert correctness • safety • evidence traceability • deployment approval

The evaluation program is broader than any single benchmark because the system has multiple failure surfaces.

Five conclusions we can support today

Each conclusion is tied to a specific body of evidence.

01 Agricultural generation cannot be treated as a purely linguistic problem.

Supported by KrishokChat v2.

02 Retrieval quality changes with farmer language and query condition.

Supported by the retrieval study.

03 Evidence binding must respect agricultural relationships, not only lexical overlap.

Supported by counterfactual and slot-ablation results.

04 The system can trade answer coverage for safer behavior instead of guessing.

Supported by retrieval-degradation and selective behavior.

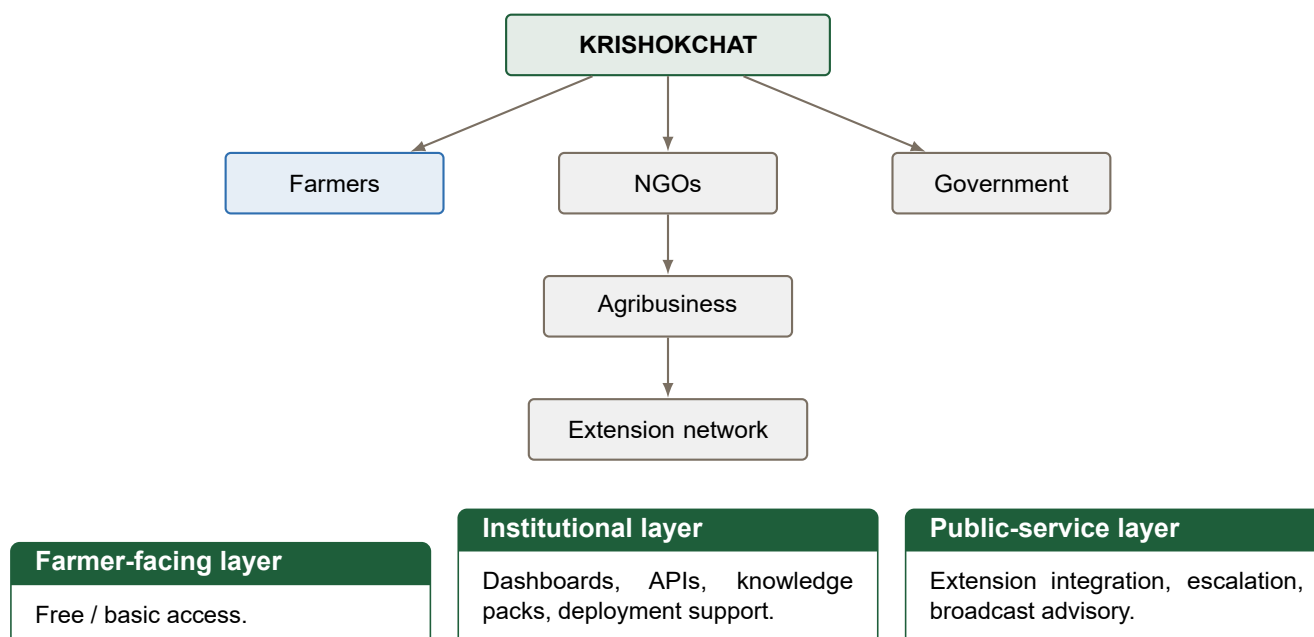
05 The application has a viable engineering path from prototype toward deployment.

Supported by codebase, deployment layers and test infrastructure — not by field adoption.

These findings led us to build KrishokChat as a platform, not a chatbot.

Designed as shared advisory infrastructure

The farmer is the beneficiary; institutions can finance deployment.



The most realistic commercialization path is not to assume that individual smallholder farmers will finance the entire service directly. KrishokChat is better positioned as shared advisory infrastructure: the farmer is the beneficiary, while institutions can finance deployment, support, knowledge integration and operational services. The same core platform can support multiple channels without creating separate products for each organization.

Production planning identifies institutional / B2B / B2G deployment as the more durable path [2].

The demand already exists; the missing layer is intelligent delivery

KrishokChat adds a digital intelligence layer to an existing ecosystem; it does not replace extension institutions.

Bangladesh does not need to invent demand for agricultural advice. Farmers already seek support through extension officers, agricultural call centres (the national helpline **16123**), weather and advisory services (BAMIS), local experts and informal information networks.

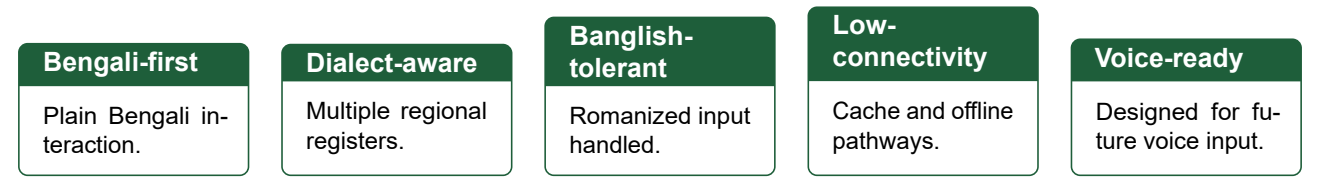
The opportunity for KrishokChat is to add a digital intelligence layer to that existing ecosystem: understanding farmer-language questions, combining text and image evidence, grounding responses in agricultural knowledge, enforcing safety checks, and routing unresolved cases toward human or institutional support. This is much stronger than pretending KrishokChat replaces existing extension institutions.

BAMIS	16123	Extension
Agricultural advisory information service.	National agricultural call center.	Field officer network (SAAOs).

Sources: existing Bangladesh advisory infrastructure [1].

Access should not depend on English, perfect spelling or reliable broadband

Inclusion is implemented as product behavior rather than a statement of intent.



Inclusion is implemented as product behavior rather than a statement of intent.

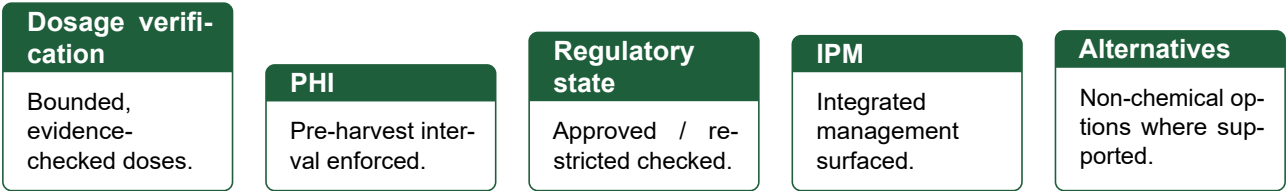
Concretely: large touch targets, plain Bengali phrasing, image-based input, progressive clarification, offline / cache pathways, and a future voice-input affordance. The frontend roadmap considers WCAG-oriented touch targets, Bengali font optimization, PWA / offline UX, on-device diagnosis and voice.

Source: frontend roadmap [2].

Better information can also mean more responsible intervention

Treatment recommendations are made conditional on evidence, supporting IPM where authoritative guidance exists.

We do not claim that KrishokChat reduces pesticide usage nationally. Instead, the system is designed to make treatment recommendations conditional on evidence and to support integrated pest management (IPM) and non-chemical alternatives where authoritative guidance exists.



This is where ethics connects to actual architecture: the same typed treatment contract that enforces safety also makes it possible to prefer lower-risk, evidence-supported interventions.

The innovation contributes directly to four development priorities

Focused SDG alignment, not a scattered checklist.

SDG 2 — Zero Hunger

Agricultural knowledge, extension, resilience.

SDG 9 — Industry & Infrastructure

Local digital infrastructure.

SDG 12 — Responsible Production

Evidence-bound intervention, safer chemical guidance.

SDG 13 — Climate Action

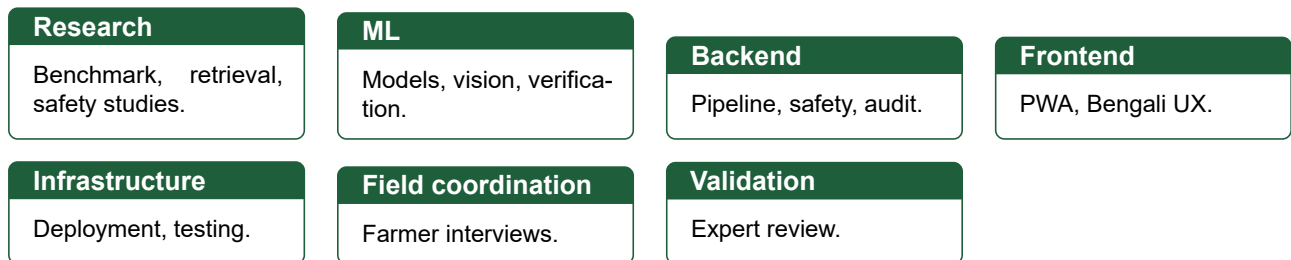
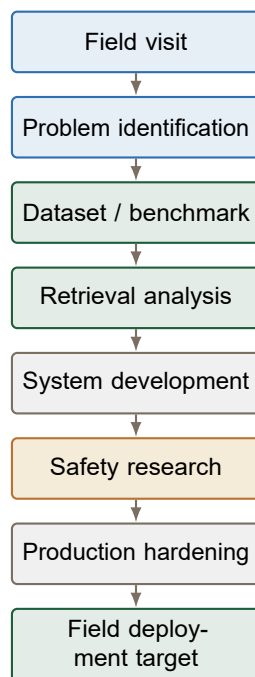
Weather-aware / proactive advisory pathways.

SDG 2 explicitly targets smallholder productivity and income, sustainable agriculture, resilience, extension and technological development. Bangladesh’s own SDG commitment connects digital agriculture to productivity, food security and the empowerment of rural farmers. KrishokChat aligns with these priorities by being a locally grounded, safety-aware agricultural intelligence layer.

Sources: UN SDG 2 targets [6]; Bangladesh SDG commitments [1].

Developed as a continuous research-to-engineering cycle

Each stage produced an artifact the next stage depended on.



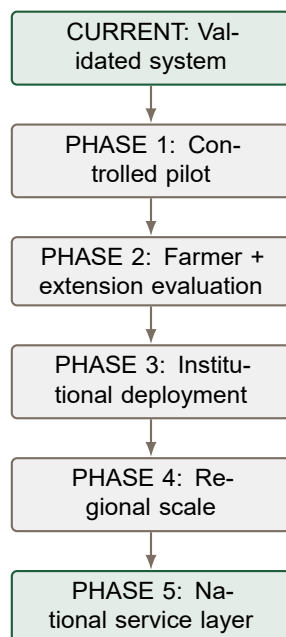
What separates KrishokChat from a feature demo

Problem-first, evidence-bound, and operationally engineered.

Typical prototype	KrishokChat
Model-first	Problem-first
One model	Multiple controlled capabilities
Answer-only	Answer + evidence + verification
Online assumed	Degraded connectivity considered
Static data	Governed evidence layer
No audit	Audit trail
No structured failure state	Clarify / refuse / escalate
One provider	Replaceable model providers
Demo code	Modular deployment-oriented architecture
Evaluation = accuracy	Safety + coverage + latency + reliability

The next step is not another prototype — it is a field deployment cycle

From validated system to national service layer, in defined phases.



Phase 1 Deploy to a controlled farmer cohort.

Phase 2 Measure expert agreement, useful coverage, safety, referral and user behavior.

Phase 3 Integrate with institutional workflows.

Phase 4 Expand crop / knowledge / service coverage.

Phase 5 Scale through public and private agricultural channels.

What support changes

The next investment reduces the gap between controlled validation and field operation.

Field validation Farmer pilots, extension evaluation, outcome measurement.	Knowledge scale Expert-reviewed agricultural knowledge packs, regulatory updates.
Deployment Hosting, edge / offline, monitoring, service reliability.	Ecosystem Institutional integration, extension workflow, partnership development.

The core system does not need to be invented from scratch. The next investment is in reducing the gap between controlled validation and field operation. Funding would support expert-reviewed knowledge expansion, farmer and extension pilots, production infrastructure, multimodal and offline deployment, operational monitoring, and institutional integration. The objective is to measure real-world value under real conditions and build the evidence required for responsible scale.

We will measure deployment by outcomes, not downloads

Success is defined by safe, correct, resolved advisory cases.

Safe advisory rate	Correct advisory rate	Appropriate abstention	Farmer task completion	Extension referral resolution

Longer-term outcome measures include crop-loss / yield outcomes, chemical-use outcomes, user trust, retention, and cost per successfully resolved case. These are field-validation targets: they define what deployment must prove, and they are deliberately stated as outcomes rather than vanity metrics.

What we know now, and what deployment must still prove

We know exactly where the evidence currently ends.

Established today

benchmark resources • retrieval findings • system architecture
• safety tests • expert review • software verification • network
simulation • local model path • production structure

Next validation

field outcomes • carrier-level delivery • farmer adoption •
extension workload • real referral follow-through • long-term
trust • yield / chemical-use impact

Evidence	Status	Meaning
v2 benchmark	Established	arXiv resource
Retrieval study	Established	arXiv study
Expert review	Measured	200 outputs
Safety attack suite	Measured	specified benchmark
Counterfactual verification	Measured	2,000 pairs
Network resilience	Simulated	emulator
GSM composition	Tested	software composition
Carrier delivery	Future	not measured
Farmer outcome	Future	field study needed
National impact	Future	not yet demonstrated

We know exactly where the evidence currently ends.

Backed by a continuous research programme

Four connected research contributions underpin the platform.

Research 01 — Provenance-Traceable Bengali Agricultural Benchmark

Contribution: knowledge / evaluation foundation. **Status:** arXiv preprint (current) [4].

Research 02 — Where Does Retrieval Fail?

Contribution: retrieval failure analysis. **Status:** arXiv:2608.14886 [5].

Research 03 — Bounded-Authority Agricultural Advisory (CEA manuscript)

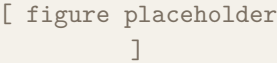
Contribution: system reliability, routing, verification, deployment architecture. **Status:** in preparation [3].

Research 04 — KrishokChat System Demonstration

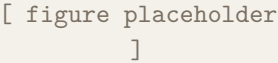
Contribution: integrated public system. **Status:** demonstration / submission.

Explore the work

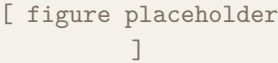
The platform and its research are openly accessible.


figures/qr_live.png

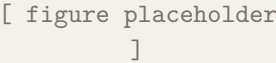
Live system
Try KrishokChat


figures/qr_github.png

Source code
GitHub repository


figures/qr_v2.png

Research
KrishokChat v2


figures/qr_retrieval.png

Retrieval study
Where Does Retrieval Fail?

Primary repository: <https://github.com/RaiyaanReza/KrishokChat-Agricultural-Advisory-System>

The repository is public and exposes backend, frontend, deployment, research artifacts, production plans and experiment materials [2]. QR placeholders swap to real codes when generated.

From field problem to trusted agricultural service

[figure placeholder]
figures/closing_system.png

KrishokChat began with a practical question from the field:

How can agricultural technology become genuinely useful to a farmer when language, evidence, uncertainty and connectivity all matter at the same time?

We answered that question by building the research foundation, measuring where current AI fails, redesigning the advisory architecture around those failures, and engineering the result into a working platform. The next step is to take that system into the field, measure its real value, and build the institutional pathway for responsible scale.

Research → Evidence → System → Field

KrishokChat is no longer only a question of whether an AI model can answer an agricultural question. The project is now about whether Bangladesh can build a locally grounded, evidence-aware, safe and deployable intelligence layer for agricultural decision support.

The next experiment is in the field.

Appendix A — Experiment catalogue (grouped by question)

Internal experiment IDs shown for traceability only.

Theme	Internal experiments
Knowledge & evidence	E19 graph traversal, E23 provenance/cache, E24 knowledge growth, E30 temporal validity, E36 source fragmentation
Routing & retrieval	E17 detection gating, E25 compact router, E32 oracle routing, E33 wrong high-confidence routing, E35 normalization, E12 retrieval degradation
Safety & verification	E02 misbinding, E03 slot ablation, E04 calibration, E05 counterfactual, E07/E08 security, E10 failure taxonomy, E11 generator robustness, E12 poisoning, E22 SMS injection
Language & multimodality	E06 dialect, E21 dialect hazard, E31 multimodal uncertainty, E37 clarification
Delivery & deployment	E09 runtime, E14 network, E15 GSM, E18 LLM dependency, E20 cost, E38 escalation, E39 IPM

Appendix B — Key empirical values

Metric	Value	Status
Core benchmark instances	85,979	Established
Real-world farmer queries	1,000	Established
Semantic knowledge units	2,946	Established
Source publications	284	Established
Hybrid RRF R@10	0.539	Measured
Counterfactual certification (typed)	0.00%	Measured
Counterfactual certification (Vanilla RAG)	72.65%	Measured
Slot ablation — dosage bounds	+31.6 pp	Measured
Retrieval degradation hazard (KrishokChat)	0.0%	Measured
Verified hazard (all generators)	0.00%	Measured
Dialect coverage (gated)	89.8 / 90.0%	Measured
Expert mean correctness	4.82 / 5	Measured
Deployment approval	96.5%	Measured
Gwet's AC1 (safety)	0.862	Measured
Cross-modal clarification trigger	100%	Measured
Deterministic advisory workload	51.04%	Measured
Zero-LLM terminal handling	61.52%	Measured
Weighted latency (tiered vs LLM)	546.15 / 1247.93 ms	Measured
Offline delivery @30% loss	58.1% vs 12.8%	Simulated
SMS injection leakage (deterministic)	0 / 1,400	Measured
Provenance mutation detection	1,000 / 1,000	Measured
Backend tests	558 pass / 8 skip / 0 fail	Measured

Appendix C — Evidence discipline

Every quantitative claim in this dossier carries an evidence label: **MEASURED** real experimental measurement; **SIMULATED** software / network simulation; **ESTABLISHED** published research resource; **TESTED** software composition; **PROJECTED** scenario or model assumption. Simulated network results are not carrier-level measurements. Field outcomes, farmer adoption and national impact are explicitly future-validation items.

References

References

- [1] Bangladesh innovation fair — about and innovator application. <https://innovationfairbd.org/about>, 2026.
- [2] Raiyaan Reza. Krishokchat agricultural advisory system. <https://github.com/RaiyaanReza/KrishokChat-Agricultural-Advisory-System>, 2026.
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